Communications Physics

https://doi.org/10.1038/s42005-026-02763-9

Article in Press

Design of topological thermal diffusion in the quasi ballistic phonon regime

Received: 5 February 2026

Accepted: 6 July 2026

Published online: 18 July 2026

Cite this article as: Funayama, K., Akura, Y., Tanaka, H. *et al.* Design of topological thermal diffusion in the quasi ballistic phonon regime. *Commun Phys* (2026). https://doi.org/10.1038/s42005-026-02763-9

Keita Funayama, Yuki Akura, Hiroya Tanaka & Jun Hirotani

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

ARTICLE IN PRESS

Design of Topological Thermal Diffusion in the Quasi Ballistic Phonon Regime

Keita Funayama[1*], Yuki Akura[2], Hiroya Tanaka[1], Jun Hirotani[2]

[1]Toyota Central R&D Laboratories, Inc., 41-1 Yokomichi, Nagakute, 480-1192, Aichi, Japan.
[2]Department of Micro Engineering, Kyoto University, Kyotodaigaku-Katsura C3, Nishikyo-ku, 615-8540, Kyoto, Japan.

*Corresponding author(s). E-mail(s): funayama@mosk.tytlabs.co.jp;

ARTICLE IN PRESS

Abstract

Thermal management at the nanoscale is critical for advanced semiconductor and quantum technologies. Topological diffusion enables robust and unidirectional heat transport, but its application to nanoscale systems, where quasi-ballistic phonon transport dominates, remains largely unexplored. Here, we demonstrate a design principle for topological thermal diffusion in one-dimensional silicon nanostructures based on the Su-Schrieffer-Heeger model. By incorporating size-dependent effective thermal conductivity into a diffusion-based topological framework, we account for nanoscale phonon transport within an effective model. The predicted decay rates agree well with experimentally measured decay rates of strongly localized topological interface states, validating the model as an effective description of nanoscale thermal transport through parameterized thermal conductivity. Our results establish a practical framework for engineering topological heat transport in nanoscale systems which include semiconductor-based quantum and thermoelectric devices, providing a unified and scalable framework for topological thermal management across diverse scales.

1 Introduction

Thermal management in solid nanostructures has been attracting significant attention across diverse fields such as semiconductor devices [1, 2], microscale thermoelectric applications [3–5], and thermal noise suppression and guiding for quantum devices [6, 7]. In nanoscale structures, thermal conduction is predominantly carried

ARTICLE IN PRESS

by phonons, which are the quantized lattice vibrations. The thermal transport mechanism via phonons depends on the size of the structures and the mean free paths (MFPs) of the phonons. Diffusive phonons dominate heat conduction when the size of the structures is much larger than the phonon MFP. In contrast, ballistic phonons contribute significantly to thermal transport when the structural scale is comparable to or smaller than the MFP. Most nanoscale solid devices consist of semiconductors; therefore, investigating phonon transport in semiconductor materials is important for the development of thermal management technologies. Phonons exhibit ballistic behavior in many materials at low temperatures, while silicon which is widely used in practical applications has a phonon MFP longer than one micrometer at room temperature [8]. The long MFP increases the number of surface scattering events in silicon structures with characteristic lengths smaller than one micrometer, resulting in a reduction of the effective thermal conductivity [9, 10]. As one of promising tools to control ballistic phonon behavior, one- and two-dimensional (1D and 2D) phononic crystals (PnCs) at the sub-micron scale have been investigated [11–15], although diffusive phonons still remain in such small structures. Therefore, heat is conducted by both diffusive and ballistic phonons in actual systems, which is referred to as quasi-ballistic phonon transport.

Quasi-ballistic phonons exhibit both wave-like and diffusive characteristics, making their thermal transport behavior fundamentally complex. This dual nature limits the applicability of Fourier-law-based descriptions and typically requires computationally expensive approaches, such as Monte Carlo simulations and first-principles calculations, to accurately evaluate thermal conductivity [8, 11]. As a result, establishing a simple and predictive design framework for thermal transport in the quasi-ballistic regime remains a significant challenge.

The topological nature has been reported to control both wave [16–33] and diffusive [34–42] phenomena. In particular, topological diffusion provides us unconventional design strategies for accurate thermal decay and anisotropic heat transfer. As a pioneer study, Hu et al. [34] demonstrated that the SSH model-structure, which was designed based on diffusion equation, showed the robust thermal decay with immunity to structural perturbations. Then, Wu and et al. [36] and Fukui [40] et al. expanded the topological diffusion to the 2D structures, and showed the topologically protected thermal decay at higher-order topological modes. Further, our previous studies reported that the 2D honeycomb lattices have a potential to realize unidirectional thermal flow for exciting topological thermal edge modes [38, 42]. However, as argued above, the diffusion-based topological model has struggled to estimate the thermal characteristics in nanoscale structures because the thermal conductivity in the diffusion equation, which has been a constant parameter in Fourier-law, varies under the influence of ballistic phonons and surface scattering. Thus, so far, the verification and scalability of topological thermal diffusion models in micro/nanostructures on the quasi-ballistic regime have been largely unexplored.

Here, we experimentally demonstrate that topological diffusion-based thermal transport and decay are well designed by introducing effective thermal conductivity k_{eff} including phonon dynamics other than diffusive phenomenon as size effects in 1D silicon nanostructures based on one of the most well-known topological models,

ARTICLE IN PRESS

Su-Schrieffer-Heeger (SSH) model. The 1D periodic structures consisted of circular sites and fine beams were excited by heating a circular site at the topological interface using laser irradiation. Using micro thermoreflectance (μ-TR), we evaluated the thermal decay rate and estimated the effective thermal conductivities k_{eff} of the topological states through temperature measurements. Our fabricated structures showed a dependence of the fine beam size on k_{eff}, indicating the contribution of ballistic and surface scattering phonons to thermal transport. Calculating the decay rate for the topological interface states based on the scalable topological diffusion model, the theoretical results are in good agreement with the experimental decay rate. Accordingly, these results indicate that topological diffusion model can be a good framework to engineer thermal transport in the quasi-ballistic regime by putting the effect of ballistic phonon dynamics into effective thermal conductivity. Furthermore, topological states allow k_{eff} to be estimated easily by fitting the theoretical decay rate to the experimental decay rate. We are confident that our research provides a comprehensive scheme for designing thermal transport from the conventional macroscale to the developing nanoscale in terms of topological diffusion model.

2 Results and Discussion

2.1 Silicon-based periodic structure and experimental setups

The silicon nanoscale structure is a 1D periodic array of unit cells consisting of two circular sub-lattices connected by fine beams, as shown in Fig. 1a. The details of the fabrication process are described in Methods section. The two circular sub-lattices, each with a radius of $r = 2.5$ μm, were connected alternately by fine beams of lengths of l_1 and l_2 with a width w, as indicated by the red square in the inset of Fig. 1a. Thus, the center-to-center distances between the sub-lattices are expressed as $L_1 = 2r + l_1$ and $L_2 = 2r + l_2$. The sub-lattices have a function as transducers from pump laser (blue line in Fig. 1a) to heat by gold (Au) and chromium (Cr) thin films of thicknesses 50 nm and 6 nm, respectively, i.e., a total thickness is $h_{\text{Au}} = 56$ nm. In our structures, the site numbered $n = 8$ in Fig. 1a was heated by the pump laser. We prepared nanoscale silicon structures with thicknesses of $h_{\text{Si}} = 0.07,\ 0.17,\ 1.0$ μm to investigate thickness-dependent thermal conductivity. In addition to thickness, the fine beams length- and width-dependent thermal conductivities were measured by using the structures with $(l_1,\ l_2) = (10,\ 5),\ (4,\ 2),\ (2,\ 1),\ (1,\ 0.5),\ (0.4,\ 0.2)$ μm and $w \approx 0.2,\ 0.5,\ 1.0,\ 1.5$ μm.

The temperature of the heated sub-lattice was measured using a μ-TR microscope with a 515 nm probe laser (green lines in Fig. 1a). Figure 1b illustrates the experimental setup of μ-TR, described in detail in our previous study [43]. A pump laser with a wavelength of 488 nm was modulated according to the pulse signal from a function generator to heat the Au/Cr film transducer. The reflectance of the probe laser changes with the temperature of the circular sub-lattice. Thus, the temporal temperature on the sub-lattice was obtained from the reflected probe laser. To reduce inherent common-mode noise in the probe laser, the reflected and reference probe laser were detected using a balanced photo detector. Then, the output signal from the oscilloscope was then Fourier-transformed, and redundant bandwidth ranges were eliminated

to improve the signal-to-noise ratio. After applying the inverse Fourier transformation, a low-noise temporal temperature profile was obtained from the probe signal. Before performing the thermal conduction measurements, we confirmed that each laser was accurately focused on a circular sub-lattice (Fig. 1b).

Figure 1c shows the time-dependent temperature signal s_T, i.e., the thermal decay curve in one pulse period, of site $n = 8$ in the structure with $(l_1,\ l_2) = (4,\ 2)$ μm, $w \approx 0.2$ μm, and $h_{Si} \approx 0.17$ μm. In our experiments, the pulse laser frequency was set to 10 kHz with a 50% duty ratio. Therefore, the site was heated by the pump laser between 0 and 50 μs (green shaded region in Fig. 1c). We observed that the temperature on the site exponentially decayed between 50 and 100 μs. A thermal decay rate α_{Exp} is expressed as $s_T(t) = s_0 \exp(-\alpha_{Exp} t)$, where s_0 is the maximum temperature signal value in the decay range. Thus, we could extract α_{Exp} by numerically fitting $\log(s_T(t))$ (red dashed line) to the experimental results (gray solid line), as shown in the inset of Fig. 1c. α_{Exp} is an average value obtained from six measurements in each structure, and

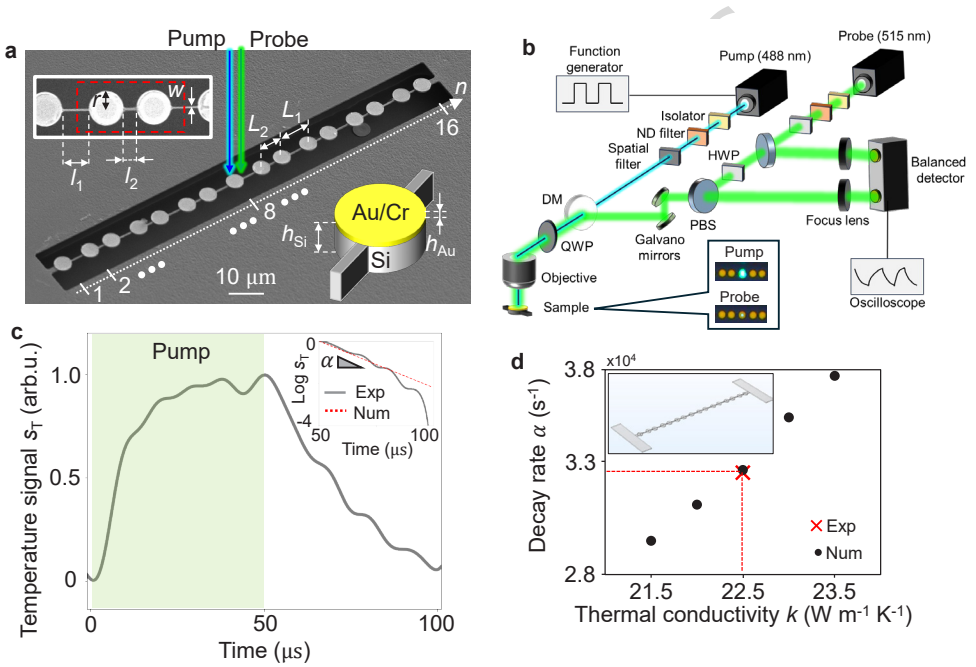


Fig. 1 1D SSH model-based topological phononic crystal for thermal transport and experimental setup. a Scanning electron microscope (SEM) image of the suspended 1D silicon nanoscale structure consisting of 16 circular sites connected by fine beams. The inset of the SEM image highlights the unit cell with a red dashed line. The schematic inset shows the structure of a circular sub-lattice. **b** Schematic of the experimental setup for μ-thermoreflectance. Here, ND filter: Neutral density filter, DM: Dichroic mirror, Q(H)WP: Quarter (Half) wave plate, and PBS: Polarizing beam splitter. **c** Time-dependent probe signal intensity s_T under 10 kHz square wave. The inset shows the logarithmic thermal decay curves between 50 and 100 μs. **d** Thermal decay rate as a function of material thermal conductivity k. The effective thermal conductivity k_{eff} is extracted by fitting the numerical decay rate (black circles) to the experimental decay rate (red crosses). The inset shows the numerical structure constructed to mimic the experimental device.

ARTICLE IN PRESS

the error bars in this study indicate the standard deviation ($\pm 1\sigma$). Using COMSOL Multiphysics, the material thermal conductivity k was treated as a free parameter to fit the numerical α (α_{Num}) to α_{Exp}, allowing the determination of the effective thermal conductivity k_{eff} of the silicon structure [8, 10], which includes the effects of ballistic and surface scattering phonons.

Figure 1d shows α_{Num} as a function of k (black circles) calculated from the structure model (inset of Fig. 1d) similar to the experimental device. We note that the actual temperature rise in the experiments is estimated from μ-TR signals to be on the order of 10 K. Thus, for the numerical simulations, we investigated the temporal temperature on site of $n = 8$ having initial temperature of $T_{\mathrm{in}} = 308.15$ K. The remaining structures were given an initial temperature of $T_0 = 298.15$ K. More details of the numerical simulation are provided in Methods section. In the case shown in Fig. 1d, the numerical result was in good agreement with the experimental result (red cross) for $k = 22.5$ W m^{-1} K^{-1}, therefore, we determined $k_{\mathrm{eff}} = 22.5$ W m^{-1} K^{-1}. All measurements were performed at room temperature under atmospheric pressure.

We accounted for the effects of thermal transport by radiation and air convection, and confirmed that thermal conduction through the silicon structures was dominant (see Supplementary Note 1 for details). In nanoscale structures, the heat transfer coefficient via convection has been enhanced to $\sim 10^3$ Wm^{-2}K^{-1} order, and thermal emissivity has been reported to be size-dependent [44–47]. As the theoretical estimation of heat transfer contribution, we note that k_{eff} is treated as an effective parameter that primarily reflects thermal conduction in the Si nano-beams, while also incorporating additional effects such as convection, interfacial resistance, boundary conditions, and fabrication-related variations. On the other hand, thermal conduction through Si nano-beams is expected to change significantly rather than a change of convection when we vary the size of the beams with fixing the size of the circular sites and measurement conditions. Through the investigation of the chance of thermal conduction in the Si nano-beams, we can estimate the contribution of the thermal conduction in effective parameter k_{eff}. Accordingly, in the next sub-section, we investigate a dependence of Si nano-beam size of effective thermal conductivity k_{eff}.

2.2 Structure Size-Dependence of the Effective Thermal Conductivity

We clarify the contribution of quasi-ballistic and surface scatted phonons in the effective thermal conductivity k_{eff} through the measurements of size-dependent thermal conduction in the Si nano-beams. In particular, we systematically investigated the effect of structural parameters, i.e., h_{Si}, $l_{1(2)}$, and w on k_{eff}. Figure 2a shows the dependence of h_{Si} on k_{eff} for $w \approx 0.2$ μm. We prepared the silicon layers with 1.0, 0.17, and 0.07 μm thicknesses. The thicknesses were selected to reveal the effect of surface phonon scattering and to make the difference in effective thermal conductivity observable, rather than to obtain a precise functional fit. In particular, the thickness of 1.0 μm structure approximate the diffusion domination with referring the previous study [8], while the thinner structures (0.17 and 0.07 μm) are expected to visualize the influence of quasi-ballistic phonon transport along the thickness direction through a change of k_{eff}. Confirmation of the silicon layers thickness is in Supplementary Note 2. We saw

ARTICLE IN PRESS

that k_{eff} decreased with decreasing h_{Si} regardless of the fine beam length. The thickness dependence of k_{eff} is clear evidence of quasi-ballistic phonon transport. Such a decrease in k_{eff} indicates that ballistic phonons scatter at the top and bottom surfaces of the fine beams. On the other hand, when h_{Si} is sufficiently large, phonon scattering in the bulk, e.g., umklapp and defect scattering, becomes dominant [48, 49]. In such large structures, the effects of surface scattering on thermal conduction are negligible, thereby, the material thermal conductivity k in Fourier-law-based pure diffusion has no dependence on thickness of the structures. Indeed, finite element method (FEM) based numerical results indicate identical decay curves regardless of h_{Si} as shown in Fig. 2b, because FEM does not account for quantum phenomena such as the surface scattering of ballistic phonon behavior. Based on the conventional thermal diffusion equation and heat flux, the thickness independence in our structure can be clearly derived in Methods section 4.3. The difference between the experimental and numerical results indicates the existence of the surface phonon scattering in the experimental devices with the range of these thicknesses. Under the conditions where all structural parameters other than h_{Si} are fixed and heat transport is dominated by conduction within the Si nano-beams, the material thermal conductivity k is treated as the only adjustable parameter for determining k_{eff} by matching the numerical results to the experimental data in Fig. 2a. As discussed above, the variation in k_{eff} dominantly originates from various phonon effects including ballistic phonon behavior in structures smaller than the phonon MFP. Therefore, we can evaluate the thermal behavior in the nanoscale structures as effective diffusion systems by utilizing k_{eff} including the complex phonon dynamics other than diffusion phenomenon.

Figure 2c shows k_{eff} as a function of l_1 for $h_{\mathrm{Si}} = 0.17$ µm. Effective thermal conductivity k_{eff} decreases with decreasing fine beam length l_1. The decrease in k_{eff} indicates the quasi-ballistic phonon-based thermal transport along the longitudinal direction of the fine beams. The quasi-ballistic phonon-based k_{eff} is proportional to $\sim {l_1}^R$ $(0 < R < 1)$, where R is a scaling factor of ballistic phonon [12, 15]. The thermal transport becomes fully diffusive and fully ballistic for $R = 0$ and 1, respectively. In our results, the nanoscale structures with each w showed R from 0.057 to 0.069. Note that the fitting lines were obtained using experimental results with $l_1 \leqq 2$ µm, because the experimental k_{eff} saturates for $l_1 > 2$ µm, indicating that l_1 exceeds the MFP of ballistic phonons along the longitudinal direction of the beam.

Finally, we investigated the dependence of k_{eff} on w as shown in Fig. 2d. We found that k_{eff} decreased with decreasing w for the structures with l_1. As w decreases, the side-surface scattering event of the ballistic phonons increases and prevents the ballistic phonons from functioning as a thermal transport carrier. Comparing with the reported long MFP of phonons in silicon at room temperature[8], the characteristic dimensions, such as beam length, width, and thickness, are comparable to or smaller than the scales of the MFP, suggesting that boundary scattering and ballistic phonon transport play an important role. While other effects incorporated in k_{eff}, such as interfacial resistance, boundary conditions, and fabrication-related variations, are difficult to quantitatively separate, the experimental results, which show the dependence of k_{eff} on the structure sizes (Fig. 2), provide indirect evidence of quasi-ballistic transport. Indeed, the experimental results in Fig. 2 demonstrate that k_{eff} exhibits a systematic

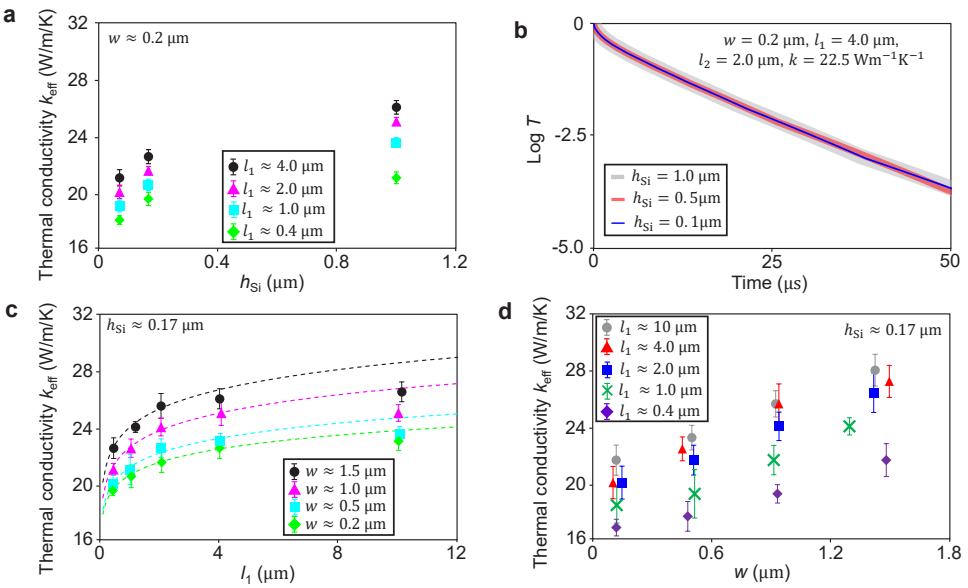


Fig. 2 Size-dependent effective thermal conductivity of the 1D-silicon nanoscale structure. a Effective thermal conductivity k_{eff} as a function of thickness of the silicon layer h_{Si}. Width of the fine beams w is fixed at approximately 0.2 μm. The error bars in this study indicate the standard deviation ($\pm 1\sigma$) from the six repetitions of the measurements. **b** Logarithmic time-dependent temperatures in the numerical models with three different thicknesses $h_{\text{Si}} = 1.0,\ 0.5,\ 0.1$ μm. The structural parameters other than h_{Si} are fixed. **c** Effective thermal conductivity k_{eff} with the error bars as a function of length of long fine beams l_1. The dashed lines are proportional to l_1^R and fitted to the experimental results with each w. **d** Effective thermal conductivity k_{eff} with the error bars as a function of width of the fine beams w. In both (**c**) and (**d**), the thickness of the silicon structure h_{Si} is fixed at approximately 0.17 μm.

size dependence that is significantly larger than the experimental error bars. This indicates that k_{eff} captures the underlying physical trends of thermal transport in the Si nano-beams, beyond serving as a simple fitting parameter. Based on these comprehensive measurements of the size-dependent k_{eff}, we verified the contributions of ballistic phonon as well as diffusive phonon to thermal transport through the silicon structures.

2.3 Topological diffusion for quasi-ballistic regime

From here, based on these experimentally observed size-dependent properties such as surface phonon scattering and partial ballistic phonon, we provide the expanded topological diffusion model, and analyze the thermal transfer in the nanoscale structure in terms of topological diffusion nature utilizing the thermal transfer devices. While the phonon dynamics in the quasi-ballistic regime vary the thermal transport properties leading to deviations from predicted eigenmodes based on a conventional diffusion, we describe the shifted eigenmodes within the diffusion-based topological model by incorporating the size-dependent phonon effects into thermal conductivity, k_{eff}. Here,

ARTICLE IN PRESS

we emphasize that the topological state for the devices was not designed from a quasi-ballistic phonon but based on diffusion , thereby, the model discussed in this study comprehensively packaged the effect of partial ballistic phonon in k_{eff}, whereas the model does not quantitatively analyze the behaviors of such phonon transport.

Our nanoscale silicon structure consists of an array of unit cells with two sublattices, A_j and B_j, as shown in Fig. 3a, where j is the unit number. As preparation for theoretical modeling, we define heat flow from and to site A_j (B_j) as $Q_{\mathrm{A}_j,\mathrm{B}_j}$ and $Q_{\mathrm{B}_{j-1},\mathrm{A}_j}$ ($Q_{\mathrm{B}_j,\mathrm{A}_{j+1}}$ and $Q_{\mathrm{A}_j,\mathrm{B}_j}$) respectively. For $Q_{\mathrm{A}_j,\mathrm{B}_j}$, the heat flow is expressed as $Q_{\mathrm{A}_j,\mathrm{B}_j} = -k_{\mathrm{eff}} S(T_{\mathrm{B}_j} - T_{\mathrm{A}_j})/L_2$, where S is the cross-sectional area of the beam, i.e., $S = wh_{\mathrm{Si}}$. Based on the heat flow and infinite periodicity with Bloch's theorem, we consider the temperature time derivatives of the temperatures $T_{\mathrm{A_j}}$ and $T_{\mathrm{B_j}}$ at each unit cell as follows:

$$\frac{\partial}{\partial t}\begin{bmatrix} T_{\mathrm{A}_j} \\ T_{\mathrm{B}_j} \end{bmatrix} = -\begin{bmatrix} D_1 + D_2 & -(D_1 e^{-i\beta} + D_2) \\ -(D_1 e^{i\beta} + D_2) & D_1 + D_2 \end{bmatrix}\begin{bmatrix} T_{\mathrm{A}_j} \\ T_{\mathrm{B}_j} \end{bmatrix}. \tag{1}$$

Here, β is the wave number vector, and $D_{1(2)}$ is the effective thermal diffusivity between B_{j-1} and A_j (A_j and B_j), i.e., $D_{1(2)} = k_{\mathrm{eff}} w/c\rho\pi r^2 L_{1(2)}$, where c (= 700 J kg^{-1} K^{-1}) and ρ (= 2329 kg m^{-3}) are the heat capacity and mass density of crystalline silicon, respectively (see the Methods section for details of the derivation). While Eq. (1) is based on diffusive phonons, the effects of ballistic phonons can modify k_{eff}, i.e., $k_{\mathrm{eff}} = k(w, l_{1,2}, h_{\mathrm{Si}})$. In the SSH model, the topology of the structure is designed by changing the lengths of inter- and intra-beams in unit cell. The topological phase transition is evaluated via the trajectories of the endpoints of the pseudospin vector $\mathbf{P} = P_{\mathrm{x}}\mathbf{x} + P_{\mathrm{y}}\mathbf{y}$, where P_{x} and P_{y} are derived as $P_{\mathrm{x}} = D_1 + D_2 \cos\theta$ and $P_{\mathrm{y}} = D_2 \sin\theta$, respectively, based on the Bloch function and Pauli matrices [34, 50, 51]. Varying θ from 0 to 2π, the trajectory of endpoint $\mathbf{P}$ describes a counterclockwise ellipse with circular (square) markers when the inter- and intra-beam lengths are l_1 (l_2) and l_2 (l_1), respectively, as shown in Fig. 3b. The winding numbers corresponding to loops around the origin are 1 and 0 for unit cells with inter-beam length of l_1 (topological) and l_2 (trivial), respectively, resulting in the topological phase transition by change of a Zak phase from 2π to 0. Note that the center of the ellipse for the topological phase shifts to the region at $P_{\mathrm{x}} > 0$ because the asymmetry between L_1 and L_2 is small due to the effect of r.

We prepared a supercell consisting of 8-unit cells to excite the topological interface state between the topological ($1 \leq j \leq 4$) and trivial ($4 < j \leq 8$) unit cells. Similar to Eq. (1), we considered the temperature-time derivative of each sub-lattice in the supercell and obtained the eigenequation $\partial\mathbf{T}/\partial t = -i\mathcal{H}\mathbf{T}$, where $\mathcal{H}$ is the following

ARTICLE IN PRESS

ARTICLE IN PRESS

Hamiltonian:

$$\mathcal{H} = -i \begin{bmatrix} D_1 & -D_1 & 0 & \cdots & \cdots & \cdots & \cdots & 0 \\ -D_2 & D_1+D_2 & -D_1 & 0 & \cdots & \cdots & \cdots & \vdots \\ \vdots & \cdots & \ddots & \cdots & \cdots & \cdots & \cdots & \vdots \\ 0 & \cdots & -D_2 & D_1+D_2 & -D_1 & \cdots & \cdots & \vdots \\ 0 & \cdots & \cdots & -D_1 & 2D_1 & -D_1 & \cdots & \vdots \\ 0 & \cdots & \cdots & \cdots & -D_1 & D_1+D_2 & -D_2 & \vdots \\ \vdots & \cdots & \cdots & \cdots & \cdots & \cdots & \ddots & \vdots \\ 0 & \cdots & \cdots & \cdots & \cdots & \cdots & -D_2 & D_2 \end{bmatrix}. \quad (2)$$

Here, $\mathbf{T} = \left[T_{\mathrm{A}_1}\ T_{\mathrm{B}_1} \cdots T_{\mathrm{A}_j}\ T_{\mathrm{B}_j} \cdots T_{\mathrm{A}_8}\ T_{\mathrm{B}_8}\ \right]$ represents the eigenfunction as the temperature field in the structure. By solving the diffusion-based eigenequation, we obtained the bulk band diagram of the supercell as shown in Fig. 3c. To calculate the band diagram, we substituted $l_1 = 4.0$, $l_2 = 2.0$, $w = 0.2$, $h_{\mathrm{Si}} = 0.17$, $r = 2.5$ µm, and $k_{\mathrm{eff}} = 22.5$ W m^{-1} K^{-1} into the Hamiltonian in Eq. (2). The eigenmodes, i.e., eigenvalues, in the bulk band diagram obtained from topological diffusion model represent the decay rate. The bulk band diagram exhibits a decay rate gap between eigenmodes $m = 8$ and 10. The circled eigenmode ($m = 9$) within this gap corresponds to a topological interfacial mode. Figure 3d shows the mode profile of $\mathbf{T}$ for eigenmode $m = 9$. We see that the mode profile is localized around the domain wall, with the dominant component at site $n = 8$, i.e., sub-lattice A_4. Accordingly, by irradiating the pump laser at site $n = 8$, the topological interface mode is selectively excited. When the interface state is well localized, the thermal response is dominated by this single mode, resulting in a stable decay behavior and good agreement with the theoretically predicted decay rate.

The next step in this study is to verify whether the scalable topological diffusion model enables to describe the experimentally obtained thermal decay rate in our nanoscale structures. Figure 4a presents the experimental decay rates (filled symbols) and theoretical decay rates (open symbols, α_{The}) for the nanoscale structures corresponding to the topological interface states. Note that we calculated α_{The} using the same structural parameters and k_{eff} as in the experimental devices. The standard deviation of α_{Exp}, become approximately the range from $\sim 10^3$ to 1×10^4, which are smaller than the size of symbols in Fig. 4a, thereby, the error bars are omitted here. We found that α_{The} was in good agreement with α_{Exp} for $l_1 > 2$ µm regardless of w. On the other hand, for $l_1 \leq 2$ µm, α_{The} deviates from α_{Exp}. To quantify this discrepancy, we define $\Delta = |\alpha_{\mathrm{Exp}} - \alpha_{\mathrm{The}}|$. Figure 4b shows Δ as a function of l_1 for each w. We found that Δ remains small for $l_1 > 2$ µm, independent of h_{Si} and w (see Supplementary Note 3), whereas Δ exponentially increases as l_1 decreases. As shown in Figs. 2a and 2d, complex phonon dynamics such as surface scattering appear by change of h_{Si} and w, thereby, these results indicate that the topological diffusion model enables evaluation of thermal decay behavior in the quasi-ballistic phonon regime. In other

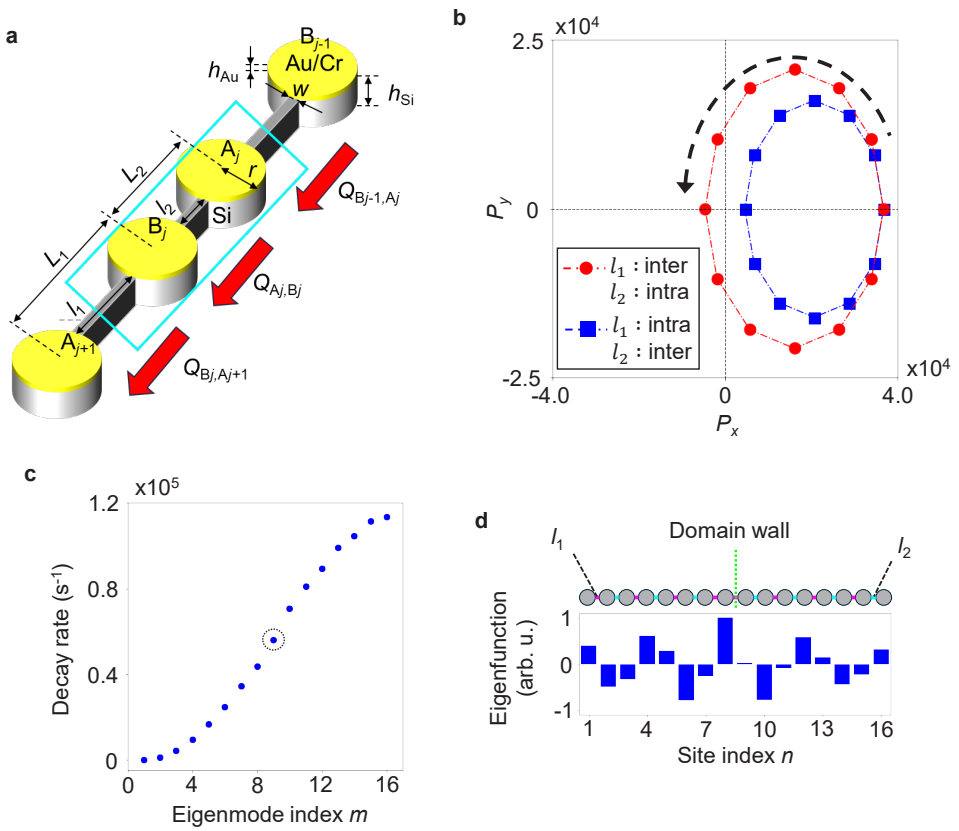


Fig. 3 Topological phase transition and interface state. a Schematic of thermal conduction in a 1D topological structure. **b** Trajectories of the endpoints of the pseudospin vector $\mathbf{P} = P_x\mathbf{x}+P_y\mathbf{y}$. The winding number corresponds to the number of loops formed by the trajectories of $\mathbf{P}$. The arrow represents the direction of the evolution of $\mathbf{P}$. Red circles and blue squares indicate the trajectories of the topological and trivial unit cells, respectively. **c** Retrieved spectrum of decay rates obtained by solving the eigen-equation with the Hamiltonian in Eq. (2) for $l_1 = 4$, $l_2 = 2$, $w = 0.2$, $r = 2.5$, and $h_{Si} = 0.17$ μm. **d** Theoretical mode profile for the topological interface state ($m = 9$ in (**c**)) in the 1D nanoscale SSH model-based supercell. The domain wall is located between sites $n = 8$ and 9.

words, the theoretical model has a potential to identify k_{eff} in topologically designed structures from purely diffusive to quasi-ballistic regions by fitting α_{The} to α_{Exp} using k_{eff} as a free parameter.

Regarding the remaining concerns, we attribute the increase in Δ as a function of l_1 to the inability to experimentally excite the topological edge states by heating only site $n = 8$ as the beam becomes shorter, because the localization of the mode profile is weakened. The localization of the topological interface states becomes stronger with larger asymmetries between D_1 and D_2. In our structures, the asymmetry between D_1 and D_2 arises solely from the difference between L_1 and L_2, since

ARTICLE IN PRESS

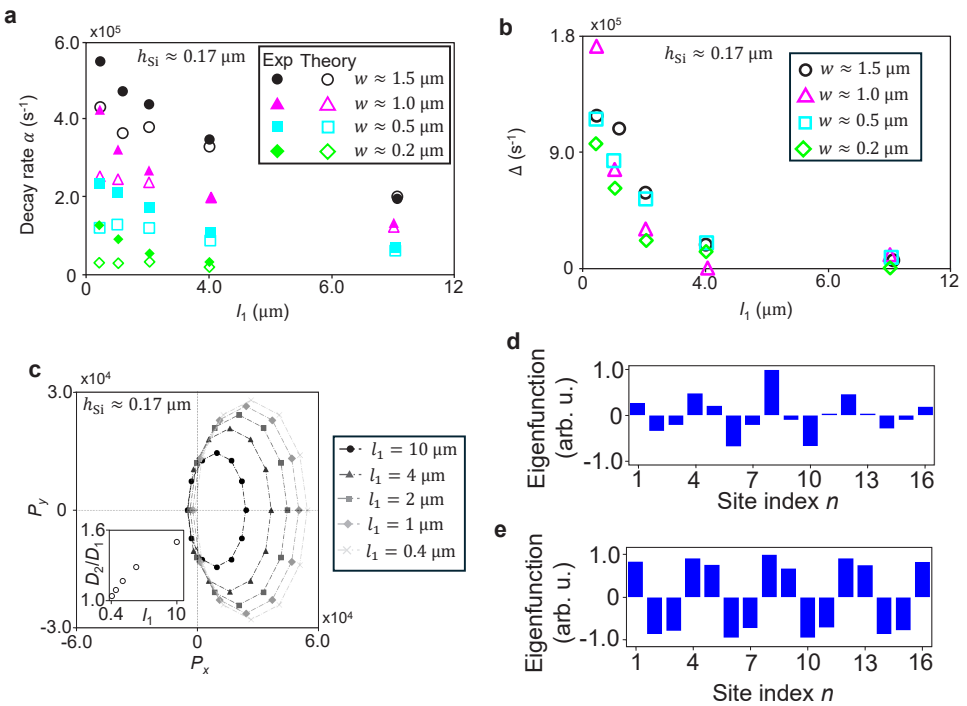


Fig. 4 Comparison of experimental and theoretical decay rates for topological interface states. a Decay rates α in the silicon nanoscale structures for topological interface states as a function of fine beam length l_1. The silicon thickness h_{Si} is fixed at approximately 0.17 μm. The experimental results were obtained by heating and measuring the temporal temperature at site $n = 8$. Note that the error bars, which are calculated as the standard deviation from the results from 6 repetitions, much smaller than the size of symbols, thereby, they are omitted. **b** Discrepancy between experimental (α_{Exp}) and theoretical (α_{The}) decay rates, defined as $\Delta = |\alpha_{Exp} - \alpha_{The}|$, as a function of fine beam length l_1. **c** Trajectories of the endpoints of the pseudospin vector $\mathbf{P} = P_x\mathbf{x} + P_y\mathbf{y}$ for topological unit cells with fine beam lengths $l_1 = 10,\ 4,\ 2,\ 1,\ 0.4$ μm. The inset shows the asymmetry ratio of the thermal diffusivity in the unit cell D_2/D_1 as a function of l_1. Theoretical mode profiles for the topological interface states ($m = 9$) for (**d**) $(l_1,\ l_2) = (10,\ 5)$ μm and (**e**) $(l_1,\ l_2) = (0.4,\ 0.2)$ μm.

the other structural parameters are identical. The center-to-center distance between the nearest-neighboring sites, $L_{1(2)}$, consists of a constant radius r and a variable beam length $l_{1(2)}$. Thus, D_1 and D_2 approach symmetry as l_1 and l_2 become relatively smaller when compared with r. Due to this reduction in asymmetry, Figure 4c shows that the center of trajectories of $\mathbf{P}$ horizontally shifted, causing the origin to approach the left edge of the ellipse. While each closed ellipse still encircles the origin, the topological property weakens as the deviation between the origin and the ellipse center increases, due to chiral symmetry breaking [52, 53]. Indeed, D_2/D_1 for $h_{Si} \approx 0.17$ μm and $w \approx 0.2$ μm approach to $D_2/D_1 = 1.0$, i.e., a symmetric structure, with decreasing beam length $l_1(l_2)$ as shown in the inset of Fig. 4c.

ARTICLE IN PRESS

Figures 4d and 4e show the theoretical profiles of the topological interface states for supercells with $(l_1,\ l_2) = (10,\ 5)$ and $(0.4,\ 0.2)$ µm, respectively. Asymmetric ratios D_2/D_1 were 1.5 and 1.04 for $(l_1,\ l_2) = (10,\ 5)$ and $(0.4,\ 0.2)$ µm, respectively, as shown in the inset of Fig. 4c. In Fig. 4d, the mode profile is localized around site $n = 8$, similar to that in Fig. 3c. Due to this localization, the topological interface state is selectively excitable by heating site $n = 8$. By contrast, in Fig. 4e, the mode profile is broadened, resembling the bulk modes. For such broadened modes, we cannot excite the topological interface states selectively by heating only site $n = 8$ because the eigenfunction T_{A_4} is not dominant in the mode profile. Accordingly, the experimentally excited mode becomes a superposition of multiple modes representing the localized profile at site $n = 8$, and the observed decay rate does not correspond to that of the single topological interface state. Consequently, we note that the decay rate α_{Exp} is extracted by fitting the measured temperature signal with a single-exponential function, while α_{Exp} should be interpreted as an effective decay rate, which represents the mixed decay rate in thermally excited modes, rather than a pure eigenmode decay. In particular, when the topological interface state is strongly localized, the measured response is dominated by a single eigenmode, resulting in good agreement with theory. On the other hand, as the localization weakens, multiple eigenmodes contribute to the measured signal, and the extracted decay rate represents an effective value obtained from a superposition of modes. Accordingly, the deviation between α_{Exp} and α_{The} can be understood as a consequence of multi-mode contributions. Indeed, α_{Exp} deviates from α_{The} as the fine beams shorten as shown in Fig. 4b. Therefore, we can reduce Δ by decreasing the size of r relative to l_1 and l_2, even for short beams.

As emphasizing again, we note that the present approach is based on an effective description of topological thermal transport, in which complex phonon dynamics in the quasi-ballistic regime are incorporated into a size-dependent effective thermal conductivity, k_{eff}. In the diffusive regime considered in previous studies (e.g., Ref. [34]), thermal transport is described by Fourier's law with a constant thermal conductivity, and topological properties emerge within this classical diffusion framework. In contrast, in the quasi-ballistic regime investigated here, phonon transport is influenced by additional effects such as boundary scattering and partial ballistic propagation, resulting in a size-dependent effective thermal conductivity. Despite these differences in transport mechanisms, our results show that the topological characteristics of the system, including the existence of localized interface states and their decay behavior, can still be captured by an effective diffusion-based model. This indicates that the concept of topological diffusion can be extended to regimes where microscopic transport mechanisms are more complex. While this framework does not explicitly resolve microscopic phonon transport mechanisms such as MFP distributions or boundary scattering, it enables a practical and predictive modeling of thermal behavior in topological diffusion systems.

Topological description based on wave-like phonon dynamics is also one of the effective approach to investigate the quasi -ballistic phonon-based thermal transfer. However, in realistic thermal transport, heat is carried by a broad distribution of phonon modes with different mean free paths and scattering processes, making it difficult to accurately describe quasi-ballistic phonon transport. In contrast, the present

ARTICLE IN PRESS

study adopts an effective approach, in which these complex phonon dynamics are incorporated into a size-dependent effective thermal conductivity within a diffusion-based topological framework. This enables a practical description of topological thermal transport in regimes where microscopic wave-based models are not directly applicable. As future works beyond the effective model, collaborating with More detailed approaches, such as Boltzmann transport equation or Monte Carlo simulations, the effective topological diffusion model could further insights into the underlying phonon physics; however, they are computationally demanding and not always suitable for device-level design, and future works because of the outside of the scope in this study.

3 Conclusions

We have investigated topological thermal diffusion behavior using the SSH model-based 1D nanoscale structures in the quasi-ballistic regime. We observed the size dependence of the effective thermal conductivity, k_{eff}, to verify the contributions of both ballistic and diffusive phonons to thermal conduction. The effective thermal conductivity k_{eff} decreases with decreasing thickness and width of the fine beams due to ballistic and surface scattering phonons. The length dependence of k_{eff} further indicates the contribution of ballistic phonons to thermal transport through the non-zero scaling factor R. By incorporating such complex phonon dynamics to effective thermal conductivity k_{eff} in the diffusion equation, we estimated thermal decay rate α at the topological interface states even in quasi-ballistic regime by using the scalable topological diffusion model. The experimental decay rate, α_{Exp}, is in good agreement with the theoretical decay rate, α_{The}, for topological interface states with localized mode profiles. This agreement demonstrates that the topological nature enables estimation of effective thermal conductivity in nanoscale structures by fitting α_{Exp} to α_{The}, using k_{eff} as a free parameter. Overall, our results open up the design scheme of thermal engineering in diverse scales through topology.

4 Methods

4.1 Numerical simulation

We used the MEMS module in COMSOL Multiphysics 6.2 to extract the effective thermal conductivity, k_{eff}, of the nanoscale structures. In the numerical simulations, the Au/Cr transducer was modeled as a thin Au layer with a total thickness of 56 nm on top of each silicon disk. All external boundaries were treated as adiabatic, neglecting heat transfer via radiation and convection, as thermal conduction through the silicon structure was assumed to be the dominant mechanism. To prevent heat accumulation within the SSH periodic structure, the silicon regions at both ends were extended to serve as thermal reservoirs with sufficiently large heat capacity. The computational domain was discretized using a fine mesh, with elements further refined in the narrow beam regions to accurately capture temperature gradients; this ensured solver convergence on the order of 10^{-8}. To replicate the experimental conditions, we set the initial temperature $T_{\text{in}} = 308.15$ K at site $n = 8$ in the numerical model shown in the inset of Fig. 1d. The initial temperature in all other regions was set to $T_0 = 298.15$ K. We

ARTICLE IN PRESS

then simulated the temporal temperature at site $n = 8$. In the simulations, the material thermal conductivities of all regions in the silicon structure were swept to fit the numerical decay rate, α_{Num}, to the experimental decay rate, α_{Exp}. We note that the thermal conductivity of the sub-lattices has a minimal effect on the decay behavior because the fine beams restrict the amount of heat transferred owing to their much smaller cross-section compared with that of the sub-lattices.

4.2 Device fabrication

We fabricated 1D nanoscale structures using a single crystal silicon-on-insulator substrate. The thicknesses of the silicon device layer, silicon dioxide (SiO_2), and silicon support layer were 1, 10, and 470 µm, respectively. The silicon device layer was etched to thicknesses of 0.17 and 0.07 µm using the Bosch process (MUC-21 ASE-Pegasus, Sumitomo Precision Products) to obtain device layers of different thicknesses. First, we patterned the circular patterns corresponding to the transducers using electron beam (EB) lithography (JEOL JBX-6300FS) with a resist (ZEP520A, Zeon Corporation). Then, the films of Au(50 nm)/Cr(6 nm) were fabricated by EB evaporation (Canon Anelva Corporation, EVD-500B) to form the thermal transducers at each site. In addition to the transducers, Au/Cr alignment patterns were simultaneously fabricated for precise alignment. The thicknesses of the silicon layer and Au thin films were measured using a stylus profiler. Next, we depicted patterns of the nanoscale silicon structures including the circular sub-lattices and fine beams with aligning their position to the transducers using alignment marks. Then, the silicon device layer was etched to form the nanoscale structures. Finally, the SiO_2 layer was removed by buffered hydrofluoric acid, and the silicon structures were suspended as bridge structures via supercritical drying.

4.3 1D SSH model-based Hamiltonian

We considered the Hamiltonian in the structure as shown in Fig. 3a. Assuming heat flows from site A_j (B_{j-1}) to B_j (A_j) as $Q_{\mathrm{A}_j,\mathrm{B}_j}$ ($Q_{\mathrm{B}_{j-1},\mathrm{A}_j}$), the temperature-time derivative of the temperature at site A_j is expressed as follows:

$$\frac{\partial T_{\mathrm{A}_j}}{\partial t} = -\frac{\Delta Q}{C} = -\frac{Q_{\mathrm{A}_j,\mathrm{B}_j} - Q_{\mathrm{B}_{j-1},\mathrm{A}_j}}{C}, \tag{3}$$

where C is the thermal capacity of the site. Further elaborating $Q_{\mathrm{A}_j,\mathrm{B}_j}$ ($Q_{\mathrm{B}_{j-1},\mathrm{A}_j}$), the heat flow is rewritten as $Q_{\mathrm{A}_j,\mathrm{B}_j(\mathrm{B}_{j-1},\mathrm{A}_j)} = -kS(T_{\mathrm{B}_j(\mathrm{A}_j)} - T_{\mathrm{A}_j(\mathrm{B}_{j-1})})/L_{2(1)}$, where $S = wh_{\mathrm{Si}}$ is the cross-section of fine beams. Considering that $C = c\rho\pi r^2 h_{\mathrm{Si}}$, where c is the specific heat, Eq. (3) is rewritten as

$$\frac{\partial T_{\mathrm{A}_j}}{\partial t} = \frac{kwh_{\mathrm{Si}}}{c\rho\pi r^2 h_{\mathrm{Si}}}\left(\frac{T_{\mathrm{B}_{j-1}}}{L_1} + \frac{T_{\mathrm{B}_j}}{L_2} - \left(\frac{1}{L_1} + \frac{1}{L_2}\right)T_{\mathrm{A}_j}\right). \tag{4}$$

In Eq. 4, we see that the thickness of silicon layer is canceled. Thus, the temporal temperature does not show a thickness dependence for a constant material thermal

ARTICLE IN PRESS

conductivity k as shown in Fig. 2b. When we normalize the thermal diffusivity to $D_{1(2)} = kw/c\rho\pi r^2 L_{1(2)}$ and consider Bloch's theorem, Eq. (4) is finalized as follows:

$$\frac{\partial T_{\mathrm{A}_j}}{\partial t} = D_1 T_{\mathrm{B}_{j-1}} + D_2 T_{\mathrm{B}_j} - (D_1 + D_2) T_{\mathrm{A}_j} = -(D_1 + D_2) T_{\mathrm{A}_j} + (D_1 e^{-i\beta} + D_2) T_{\mathrm{B}_j}. \tag{5}$$

By deriving $\partial T_{\mathrm{B}_j}/\partial t$ similarly to $\partial T_{\mathrm{A}_j}/\partial t$, we obtain the Hamiltonian shown in Eq. (1). We use python3 to theoretically calculate and analyze the experimental and numerical results in this study.

5 Data availability

Source data of this study for figures are available in figshare [https://doi.org/10.6084/m9.figshare.32788554]. Other data for this paper are available from the corresponding author upon request.

6 Acknowledgements

A part of this work was supported by Nagoya University microstructural characterization platform as a program of 'Advanced Research Infrastructure for Materials and Nanotechnology in Japan (ARIM)' of the Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan.

7 Author contributions

K.F. performed simulations, fabrications, experiments, and analysis. Y.A. performed experiments. H.T. supported the reinforcement of theoretical aspects, and analysis of simulations. J.H. contributed to discussions on theoretical and numerical analysis and supervised this project. All authors contributed to discussions and manuscript preparation.

8 Competing interests

The authors declare no competing interests.

9 Additional information

The Supporting Information is available free of charge. Details of the estimation of the dominant thermal transport phenomenon in our experimental setup, measurement of the thickness of silicon layers, and experimental and theoretical decay rates estimated using the topological scheme for several silicon thicknesses.

ARTICLE IN PRESS

ARTICLE IN PRESS

References

[1] Zhou, J., Zhong, L., Feng, X., Zhang, W., Liu, X., Zhou, H., Liu, Z., Hao, Y., Zhang, J.: Recent advances in device-level thermal management technologies for wide bandgap semiconductor: A review. IEEE Transactions on Electron Devices **72**(6), 2769–2782 (2025) https://doi.org/10.1109/TED.2025.3562506

[2] Li, M., Li, S., Zhang, Z., Su, C., Wong, B., Hu, Y.: Advancing thermal management technology for power semiconductors through materials and interface engineering. Accounts of Materials Research **6**(5), 563–576 (2025) https://doi.org/10.1021/accountsmr.4c00349

[3] Fernández-Yáñez, P., Romero, V., Armas, O., Cerretti, G.: Thermal management of thermoelectric generators for waste energy recovery. Applied Thermal Engineering **196**, 117291 (2021) https://doi.org/10.1016/j.applthermaleng.2021.117291

[4] Hochbaum, A.I., Chen, R., Delgado, R.D., Liang, W., Garnett, E.C., Najarian, M., Majumdar, A., Yang, P.: Enhanced thermoelectric performance of rough silicon nanowires. Nature **451**, 163–167 (2008) https://doi.org/10.1038/nature06381

[5] Zhang, D., Mao, Y., Bai, P., Li, Q., He, W., Cui, H., Ye, F., Li, C., Ma, R., Chen, Y.: Multifunctional superelastic graphene-based thermoelectric sponges for wearable and thermal management devices. Nano Letters **22**(8), 3417–3424 (2022) https://doi.org/10.1021/acs.nanolett.2c00696

[6] Ren, H., Matheny, M.H., MacCabe, G.S., Luo, J., Pfeifer, H., Mirhosseini, M., Painter, O.: Two-dimensional optomechanical crystal cavity with high quantum cooperativity. Nature Communications **11**(1), 3373 (2020) https://doi.org/10.1038/s41467-020-17182-9

[7] Ren, H., Shah, T., Pfeifer, H., Brendel, C., Peano, V., Marquardt, F., Painter, O.: Topological phonon transport in an optomechanical system. Nature Communications **13**(1), 3476 (2022) https://doi.org/10.1038/s41467-022-30941-0

[8] Anufriev, R., Gluchko, S., Volz, S., Nomura, M.: Quasi-ballistic heat conduction due to lévy phonon flights in silicon nanowires. ACS Nano **12**(12), 11928–11935 (2018) https://doi.org/10.1021/acsnano.8b07597

[9] Poborchii, V., Morita, Y., Hattori, J., Tada, T., Geshev, P.I.: Corrugated si nanowires with reduced thermal conductivity for wide-temperature-range thermoelectricity. Journal of Applied Physics **120**(15), 154304 (2016) https://doi.org/10.1063/1.4965302

[10] Kim, B., Barbier-Chebbah, F., Ogawara, Y., Jalabert, L., Yanagisawa, R., Anufriev, R., Nomura, M.: Anisotropy reversal of thermal conductivity in silicon

ARTICLE IN PRESS

nanowire networks driven by quasi-ballistic phonon transport. ACS Nano **18**(15), 10557–10565 (2024) https://doi.org/10.1021/acsnano.3c12767

[11] Regner, K.T., Sellan, D.P., Su, Z., Amon, C.H., McGaughey, A.J.H., Malen, J.A.: Broadband phonon mean free path contributions to thermal conductivity measured using frequency domain thermoreflectance. Nature Communications **4**(1), 1640 (2013) https://doi.org/10.1038/ncomms2630

[12] Anufriev, R., Ramiere, A., Maire, J., Nomura, M.: Heat guiding and focusing using ballistic phonon transport in phononic nanostructures. Nature Communications **8**(1), 15505 (2017) https://doi.org/10.1038/ncomms15505

[13] Verdier, M., Anufriev, R., Ramiere, A., Termentzidis, K., Lacroix, D.: Thermal conductivity of phononic membranes with aligned and staggered lattices of holes at room and low temperatures. Physical Review B **95**, 205438 (2017) https://doi.org/10.1103/PhysRevB.95.205438

[14] Wu, Y., Ordonez-Miranda, J., Gluchko, S., Anufriev, R., Meneses, D.D.S., Campo, L.D., Volz, S., Nomura, M.: Enhanced thermal conduction by surface phonon-polaritons. Science Advances **6**(40), 4461 (2020) https://doi.org/10.1126/sciadv.abb4461

[15] Maire, J., Anufriev, R., Yanagisawa, R., Ramiere, A., Volz, S., Nomura, M.: Heat conduction tuning by wave nature of phonons. Science Advances **3**(8), 1700027 (2017) https://doi.org/10.1126/sciadv.1700027

[16] Liu, Y., Chen, X., Xu, Y.: Topological phononics: From fundamental models to real materials. Advanced Functional Materials **30**(8), 1904784 (2020) https://doi.org/10.1002/adfm.201904784

[17] Ma, J., Xi, X., Sun, X.: Experimental demonstration of dual-band nanoelectromechanical valley-hall topological metamaterials. Advanced Materials **33**(10), 2006521 (2021) https://doi.org/10.1002/adma.202006521

[18] Xue, H., Yang, Y., Zhang, B.: Topological valley photonics: Physics and device applications. Advanced Photonics Research **2**(8), 2100013 (2021) https://doi.org/10.1002/adpr.202100013

[19] Hashemi, A., Zakeri, M.J., Jung, P.S., Blanco-Redondo, A.: Topological quantum photonics. APL Photonics **10**(1), 010903 (2025) https://doi.org/10.1063/5.0239265

[20] Funayama, K., Yatsugi, K., Miura, A., Iizuka, H.: Control of coupling between micromechanical topological waveguides. International Journal of Mechanical Sciences **236**, 107755 (2022) https://doi.org/10.1016/j.ijmecsci.2022.107755

ARTICLE IN PRESS

ARTICLE IN PRESS

[21] Funayama, K., Yatsugi, K., Iizuka, H.: Topological quantum photonics. Communications Physics **7**(1), 409 (2024) https://doi.org/10.1038/s42005-024-01899-w

[22] Isoniemi, T., Bouteyre, P., Hu, X., Benimetskiy, F., Wang, Y., Skolnick, M.S., Krizhanovskii, D.N., Tartakovskii, A.I.: Realization of z2 topological photonic insulators made from multilayer transition metal dichalcogenides. ACS Nano **18**(47), 32547–32555 (2024) https://doi.org/10.1021/acsnano.4c09295

[23] Miniaci, M., Pal, R.K.: Design of topological elastic waveguides. Journal of Applied Physics **130**(14), 141101 (2021) https://doi.org/10.1063/5.0057288

[24] Tang, G.-J., Chen, X.-D., Sun, L., Guo, C.-H., Li, M.-Y., Tian, Z.-T., Chen, H.-H., Wang, H.-W., Sun, Q.-Y., Pan, Y.-D., He, X.-T., Su, Y.-K., Dong, J.-W.: Broadband and fabrication-tolerant 3-db couplers with topological valley edge modes. Light: Science & Applications **13**(1), 166 (2024) https://doi.org/10.1038/s41377-024-01512-3

[25] Yu, S.-Y., He, C., Wang, Z., Liu, F.-K., Sun, X.-C., Li, Z., Lu, H.-Z., Lu, M.-H., Liu, X.-P., Chen, Y.-F.: Elastic pseudospin transport for integratable topological phononic circuits. Nature Communications **9**(1), 3072 (2018) https://doi.org/10.1038/s41467-018-05461-5

[26] He, X.-T., Liang, E.-T., Yuan, J.-J., Qiu, H.-Y., Chen, X.-D., Zhao, F.-L., Dong, J.-W.: A silicon-on-insulator slab for topological valley transport. Nature Communications **10**(1), 872 (2019) https://doi.org/10.1038/s41467-019-08881-z

[27] Redon, Q., Liu, Q., Bouhiron, J.-B., Mittal, N., Fabre, A., Lopes, R., Nascimbene, S.: Realizing the entanglement hamiltonian of a topological quantum hall system. Nature Communications **15**(1), 10086 (2024) https://doi.org/10.1038/s41467-024-54085-5

[28] Zhang, Q., Lee, D., Zheng, L., Ma, X., Meyer, S.I., He, L., Ye, H., Gong, Z., Zhen, B., Lai, K., Johnson, A.T.C.: Gigahertz topological valley hall effect in nanoelectromechanical phononic crystals. Nature Electronics **5**(3), 157 (2022) https://doi.org/10.1038/s41928-022-00732-y

[29] Wang, Z., Chong, Y., Joannopoulos, J.D., Soljačić, M.: Observation of unidirectional backscattering-immune topological electromagnetic states. Nature **461**(7265), 772–775 (2009) https://doi.org/10.1038/nature08293

[30] Cha, J., Kim, K.W., Daraio, C.: Experimental realization of on-chip topological nanoelectromechanical metamaterials. Nature **564**(7735), 229–233 (2018) https://doi.org/10.1038/s41586-018-0764-0

[31] Yoshimi, H., Yamaguchi, T., Ota, Y., Arakawa, Y., Iwamoto, S.: Slow light waveguides in topological valley photonic crystals. Optics Letters **45**(9), 2648–2651 (2020) https://doi.org/10.1364/OL.391764

ARTICLE IN PRESS

ARTICLE IN PRESS

[32] Barik, S., Karasahin, A., Flower, C., Cai, T., Miyake, H., DeGottardi, W., Hafezi, M., Waks, E.: A topological quantum optics interface. Science **359**(6376), 666–668 (2018) https://doi.org/10.1126/science.aaq0327

[33] Blanco-Redondo, A., Bell, B., Oren, D., Eggleton, B.J., Segev, M.: Topological protection of biphoton states. Science **362**(6414), 568–571 (2018) https://doi.org/10.1126/science.aau4296

[34] Hu, H., Han, S., Yang, Y., Liu, D., Xue, H., Liu, G.-G., Cheng, Z., Wang, Q.J., Zhang, S., Zhang, B., Luo, Y.: Observation of topological edge states in thermal diffusion. Advanced Materials **34**(31), 2202257 (2022) https://doi.org/10.1002/adma.202202257

[35] Qi, M., Wang, D., Cao, P.-C., Zhu, X.-F., Qiu, C.-W., Chen, H., Li, Y.: Geometric phase and localized heat diffusion. Advanced Materials **34**(32), 2202241 (2022) https://doi.org/10.1002/adma.202202241

[36] Wu, H., Hu, H., Wang, X., Xu, Z., Zhang, B., Wang, Q.J., Zheng, Y., Zhang, J., Cui, T.J., Luo, Y.: Higher-order topological states in thermal diffusion. Advanced Materials **35**(14), 2210825 (2023) https://doi.org/10.1002/adma.202210825

[37] Funayama, K., Hirotani, J., Tanaka, H.: Evaluation of topological protection in kagome lattice-based thermal diffusion systems. Applied Physics Letters **124**(23), 232201 (2024) https://doi.org/10.1063/5.0214412

[38] Funayama, K., Hirotani, J., Miura, A., Tanaka, H.: Selectable diffusion direction with topologically protected edge modess. Communications Physics **6**(1), 364 (2023) https://doi.org/10.1038/s42005-023-01490-9

[39] Liu, Z., Jin, P., Lei, M., Wang, C., Marchesoni, F., Jiang, J.-H., Huang, J.: Topological thermal transport. Nature Reviews Physics **6**(9), 554 (2024) https://doi.org/10.1038/s42254-024-00745-w

[40] Fukui, T., Yoshida, T., Hatsugai, Y.: Higher-order topological heat conduction on a lattice for detection of corner states. Physical Review E **108**, 024112 (2023) https://doi.org/10.1103/PhysRevE.108.024112

[41] Yoshida, T., Hatsugai, Y.: Bulk-edge correspondence of classical diffusion phenomena. Scientific Reports **11**, 888 (2021) https://doi.org/10.1038/s41598-020-80180-w

[42] Nakane, J.J., Funayama, K., Yamakage, A.: Theory of topological diffusion in a honeycomb lattice. Physical Review B **112**, 195408 (2025) https://doi.org/10.1103/pkmj-6h2x

[43] Akura, Y., Ikeda, Y., Matsunaga, Y., Shimofuri, M., Banerjee, A., Tsuchiya, T., Hirotani, J.: Frequency-domain thermoreflectance with beam offset without

ARTICLE IN PRESS

the spot distortion for accurate thermal conductivity measurement of anisotropic materials. Review of Scientific Instruments **96**(1), 014902 (2025) https://doi.org/10.1063/5.0237004

[44] Xu, S., Xu, Y., Zhang, J., Gao, J., Wang, X., Zhang, X., Yue, Y.: Ballistic transport enhanced heat convection at nanoscale hotspots. Journal of Applied Physics **136**(16), 164306 (2024) https://doi.org/10.1063/5.0221352

[45] Pulavarthy, R.A., Alam, M.T., Haque, M.A.: Effect of heated zone size on micro and nanoscale convective heat transfer. International Communications in Heat and Mass Transfer **52**, 56–60 (2014) https://doi.org/10.1016/j.icheatmasstransfer.2014.01.016

[46] Edalatpour, S., Francoeur, M.: Size effect on the emissivity of thin films. Journal of Quantitative Spectroscopy and Radiative Transfer **118**, 75–85 (2013) https://doi.org/10.1016/j.jqsrt.2012.12.012

[47] Ravindra, N.M., Sopori, B., Gokce, O.H., Cheng, S.X., Shenoy, A., Jin, L., Abedrabbo, S., Chen, W., Zhang, Y.: Emissivity measurements and modeling of silicon-related materials: An overview. International Journal of Thermophysics **22**, 1593–1611 (2001) https://doi.org/10.1023/A:1012869710173

[48] Gurunathan, R., Hanus, R., Dylla, M., Katre, A., Snyder, G.J.: Analytical models of phonon–point-defect scattering. Physical Review Applied **13**, 034011 (2020) https://doi.org/10.1103/PhysRevApplied.13.034011

[49] Jo, I., Pettes, M.T., Kim, J., Watanabe, K., Taniguchi, T., Yao, Z., Shi, L.: Thermal conductivity and phonon transport in suspended few-layer hexagonal boron nitride. Nano Letters **13**, 550 (2013) https://doi.org/10.1103/PhysRevApplied.13.034011

[50] Li, L., Lee, C.H., Gong, J.: Geometric characterization of non-hermitian topological systems through the singularity ring in pseudospin vector space. Physical Review B **100**, 075403 (2019) https://doi.org/10.1103/PhysRevB.100.075403

[51] Jiang, J., Ren, J., Guo, Z., Zhu, W., Long, Y., Jiang, H., Chen, H.: Seeing topological winding number and band inversion in photonic dimer chain of split-ring resonators. Physical Review B **101**, 165427 (2020) https://doi.org/10.1103/PhysRevB.101.165427

[52] Asbóth, J.K., Oroszlány, L., Pályi, A.: The Su-Schrieffer-Heeger (SSH) Model, pp. 1–22. Springer, Cham (2016). https://doi.org/10.1007/978-3-319-25607-8_1

[53] Lan, Z., Chen, M.L.N., Gao, F., Zhang, S., Sha, W.E.I.: A brief review of topological photonics in one, two, and three dimensions. Reviews in Physics **9**, 100076 (2022) https://doi.org/10.1016/j.revip.2022.100076

ARTICLE IN PRESS